Lab Assignment 7: Credit Risk Assessment

CSAI2017P: Applied Machine Learning (Lab)

Experiment 9 (Unit VI) • CO2, CO4

Instructor: Dr. Tofik Ali **Due:** announced in the lab
Student Name: ______________________ **SAP ID:** ______________
Batch: __________ **Lab quiz:** LQ7, after submission

Objective

- Build a scoring model where the two error types have very different costs.
- Use ROC and AUC correctly, and know what they hide.
- Examine a model for disparate outcomes across groups.

Datasets used in this assignment

A3 (Statlog German Credit).
Full descriptions, sources and loading snippets for all anchor datasets are in `Anchor-Datasets.pdf`, alongside this brief.

Instructions

- Use a train/test split (80/20 unless stated) and set `random_state` for reproducibility.
- Fit every transformer inside a `Pipeline` so nothing is fitted on the test set.
- Report the metric named in the question. A number without units or context earns no marks.
- Every question asks for a short written observation. Code without commentary caps you at half marks.
- The lab quiz that follows will ask you to read, trace and modify *your own* submitted code.

Submit:

- Notebook: `AML_LA07_<SAPID>.ipynb` with all cells executed and outputs visible.
- Report: 2–3 pages PDF, or a `README.md`, carrying your results table and observations.
- Do not upload datasets. Cite the source and include the loading code.

Section	Level	Choice rule	Marks
A	Easy	Attempt any 2 of 3 (2 marks each)	4
B	Medium	Attempt any 1 of 2 (4 marks each)	4
C	Hard	Compulsory	2
Total			**10**

Course outcomes assessed by this assignment: **CO2, CO4**. Attempting more than the choice rule allows is not penalised; the best attempts are marked.

Section A (Easy) — attempt any 2 of 3 $(2 \times 2 = 4)$

A1. Load and baseline **[2 marks]**
Dataset: A3 (credit-g). Target: `class`*.*

(a) Fetch the dataset, report its shape and the proportion of bad-credit cases.

(b) Fit a logistic regression in a pipeline with one-hot encoding and scaling; report accuracy.

(c) State the majority-class accuracy and compare in two lines.

A2. The cost matrix **[2 marks]**
Dataset: A3.

(a) Print the confusion matrix for your A1 model.

(b) The dataset's own documentation says lending to a bad customer costs five times as much as refusing a good one. Compute your model's total cost under that matrix.

(c) State in two lines whether accuracy was the right metric.

A3. ROC and AUC **[2 marks]**
Dataset: A3.

(a) Plot the ROC curve and report AUC.

(b) Mark the operating point that minimises the cost from A2.

(c) State in two lines what AUC measures that accuracy does not.

Section B (Medium) — attempt any 1 of 2 $(1 \times 4 = 4)$

B1. Cost-sensitive training **[4 marks]**
Dataset: A3.

(a) Retrain with `class_weight={good:1, bad:5}` and separately by tuning the decision threshold on a validation split.

(b) Report accuracy, recall on the bad class and total cost for the baseline and both approaches.

(c) State in 4–6 lines which approach you would ship and why.

B2. Group outcomes **[4 marks]**
Dataset: A3.

(a) Split the test set by a protected-style attribute available in the data (for example `personal_status` or age band).

(b) Report approval rate, recall and false-positive rate per group in one table.

(c) In 6–8 lines, state whether the differences you see are evidence of a problem, and what further information you would need to decide.

Section C (Hard) — compulsory (2)

C1. Explain one decision **[2 marks]**

Dataset: A3.

(a) Pick one applicant your model rejected and list the three features that pushed the score most.

(b) Write the two-sentence explanation you would give that applicant.

End of Lab Assignment 7

Sources: Müller and Guido, *Introduction to Machine Learning with Python*, companion notebooks.;McKinney, *Python for Data Analysis* (3e), O'Reilly, 2022.;Scikit-learn user guide, accessed 2026-08-04.;Matplotlib and Seaborn documentation, accessed 2026-08-04.